

- ❖ First law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

- ❖ Work done.

$$\Delta W = P\Delta V$$

$$\therefore \Delta Q = \Delta U + P\Delta V$$

- ❖ Relation between specific heats for a gas

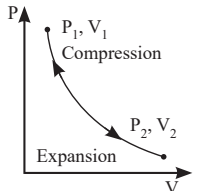
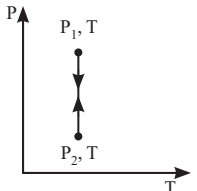
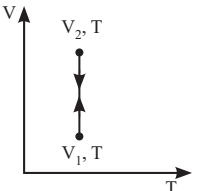
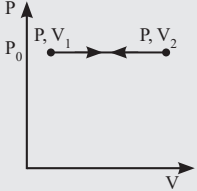
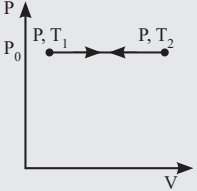
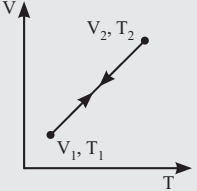
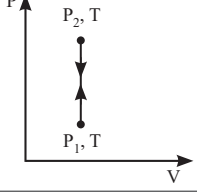
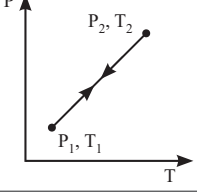
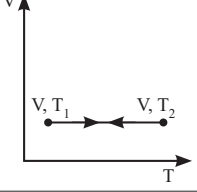
$$C_p - C_v = R$$

Polytropic Process

It is a thermodynamic process that can be expressed as follows:

$$PV^x = \text{Constant}$$

x (Polytropic exponent)	Type of standard process	Expression
0	Isobaric ($dP = 0$)	$P = \text{Constant}$
1	Isothermal ($dT = 0$)	$PV = \text{Constant}$
γ	Adiabatic ($dQ = 0$)	$PV^\gamma = \text{Constant}$
∞	Isochoric ($dV = 0$)	$V = \text{Constant}$

Process	Definition	P-V graph	P-T graph	V-T graph
Isothermal	Temperature constant			
Isobaric	Pressure constant			
Isochoric	Volume constant			

For any General Process

$$\Delta Q = \Delta U + W$$

$$\Rightarrow nC\Delta T = \frac{f}{2}nR\Delta T + \frac{nR\Delta T}{1-x}$$

[$\because$ Work done in a general polytropic process = $[nR\Delta T/(1-x)]$]

$$\Rightarrow C = \frac{f}{2}R + \frac{R}{1-x}$$

For infinitesimal changes in Q , U , and W , we can write,

$$dQ = dU + dW$$

$$\Rightarrow nCdT = \frac{f}{2}nRdT + PdV$$

$$\Rightarrow C = \frac{f}{2}R + \frac{P}{n} \frac{dV}{dT}$$

Process	Equation of State	W	ΔU
Isobaric ($dP = 0$)	$\frac{V}{T} = c$	$P(V_f - V_i) = nR(T_f - T_i)$	$\frac{f}{2}nR(T_f - T_i) = \frac{f}{2}P(V_f - V_i)$
Isochoric ($dV = 0$)	$\frac{P}{T} = c$	0	$\frac{f}{2}nR(T_f - T_i) = \frac{f}{2}V(P_f - P_i)$
Isothermal ($dT = 0$)	$PV = c$	$nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$	0
Adiabatic ($dQ = 0$)	$PV^\gamma = c$	$\frac{f}{2}nR(T_i - T_f)$ $= \frac{f}{2}(P_i V_i - P_f V_f)$	$\frac{f}{2}nR(T_f - T_i)$

Process	ΔQ
Isobaric ($dP = 0$)	$\left(\frac{f}{2} + 1\right)nR\Delta T = \left(\frac{f}{2} + 1\right)P(V_f - V_i)$
Isochoric ($dV = 0$)	$\frac{f}{2}nR(T_f - T_i) = \frac{f}{2}V(P_f - P_i)$
Isothermal ($dT = 0$)	$nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$
Adiabatic ($dQ = 0$)	0

❖ Slope of adiabatic = γ (slope of isotherm)

❖ **Carnot engine**

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$W = Q_1 - Q_2$$

$$(\text{efficiency}) \eta = \frac{W}{Q_1}$$

❖ **Refrigerator**

Coefficient of performance is β

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{Q_2}{W}$$

$$\beta = \frac{1 - \eta}{\eta}$$

❖ **Heat pump**

$$\alpha = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{1}{\eta}$$